COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S3
DATE PERFORMED:	08/13/2026	**DATE SUBMITTED:**	08/13/2026
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Exam	**Prelim Exam Skills**

1. PROCEDURE OUTPUT

Hostname	Operating System	IP Address
jppios	Ubuntu	192.168.56.101
jpbpios	Ubuntu	192.168.56.102
centos1	CentOS	

Ansible Version

– workstation

```
jbbpios@workstation:~/Desktop$ ansible --version
ansible [core 2.17.14]
  config file = None
  configured module search path = ['/home/jbbpios/.ansible/plugins/modules', '/u
sr/share/ansible/plugins/modules']
  ansible python module location = /home/jbbpios/.local/lib/python3.10/site-pack
ages/ansible
  ansible collection location = /home/jbbpios/.ansible/collections:/usr/share/an
sible/collections
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0] (/usr/bin/
python3)
  jinja version = 3.1.6
  libyaml = True
jbbpios@workstation:~/Desktop$
```

– jppios

```
jppios@server1:~/Desktop$ ansible --version
ansible [core 2.17.14]
  config file = None
  configured module search path = ['/home/jppios/.ansible/plugins/modules', '/us
r/share/ansible/plugins/modules']
  ansible python module location = /home/jppios/.local/lib/python3.10/site-packa
ges/ansible
  ansible collection location = /home/jppios/.ansible/collections:/usr/share/ans
ible/collections
  executable location = /home/jppios/.local/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0] (/usr/bin/
python3)
  jinja version = 3.1.6
  libyaml = True
jppios@server1:~/Desktop$
```

– jpbpios

```
jpbpios@server2:~/Desktop$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/jpbpios/.ansible/plugins/modules', '/u
sr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
jpbpios@server2:~/Desktop$ s
```

SSH Connectivity

```
jppios@server1: ~
jbbpios@workstation:~/Desktop$ ssh jppios@192.168.56.101
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

288 updates can be applied immediately.
208 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

3 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Jul 16 09:00:29 2026 from 192.168.56.103
jppios@server1:~$
```

```
jbbpios@workstation:~/Desktop$ ssh jpbpios@192.168.56.102
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-124-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

462 updates can be applied immediately.
385 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

3 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
Last login: Thu Jul 16 08:52:56 2026 from 192.168.56.103
jpbpios@server2:~$
```

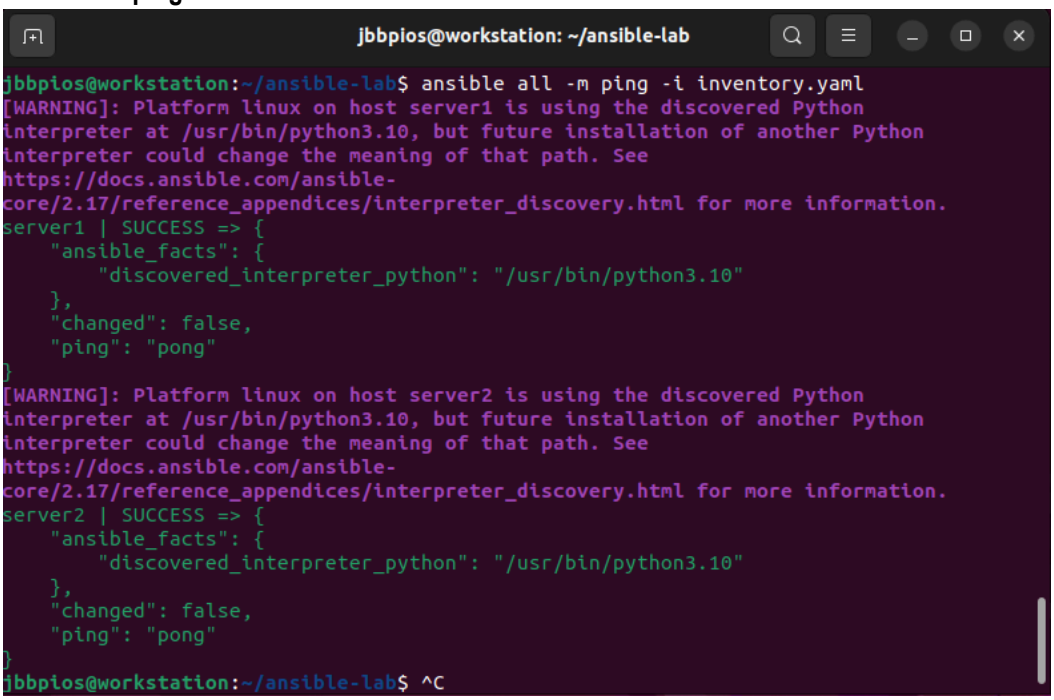
Inventory.yaml



Successful inventory verification

```
jbbpios@workstation:~/ansible-lab$ ansible all --list-hosts -i inventory.yaml
  hosts (2):
    server1
    server2
```

Successful ping results



Demonstrate successful execution of the metagroup



Demonstrate successful use of variables

```
TASK [Show Operating System] ********************************************************
ok: [server1] => {
    "msg": "Host server1 is running Ubuntu"
}
ok: [server2] => {
    "msg": "Host server2 is running Ubuntu"
}

PLAY RECAP **************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped
=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped
=0    rescued=0    ignored=0

jbbpios@workstation:~/ansible-lab$
```

Successful playbook execution

```
PLAY RECAP **************************************************************************
server1                    : ok=4    changed=0    unreachable=0    failed=0    skipped
=0    rescued=0    ignored=0
server2                    : ok=4    changed=0    unreachable=0    failed=0    skipped
=0    rescued=0    ignored=0
```

Successful task completion on all managed nodes

```
TASK [Ping all managed nodes] *******************************************************
ok: [server2]
ok: [server1]

TASK [Display Hello World] ***********************************************************
ok: [server1] => {
    "msg": "Hello World from server1"
}
ok: [server2] => {
    "msg": "Hello World from server2"
}

TASK [Display Operating System] ******************************************************
ok: [server1] => {
    "msg": "Operating System: Ubuntu"
}
ok: [server2] => {
    "msg": "Operating System: Ubuntu"
}
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a new group named servers
Create a playbook named challenge.yaml
Run the playbook and demonstrate that only hosts in the servers group receive the message.

```
jbbpios@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml challenge.yaml

PLAY [Supplementary Challenge] *************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host server1 is using the discovered Python interpreter
at /usr/bin/python3.10, but future installation of another Python interpreter could
change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [Display Welcome] *********************************************************
ok: [server1] => {
    "msg": "Welcome to Ansible Laboratory"
}

PLAY RECAP *********************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
   ignored=0
```

Guide Questions

1. What is the role of the Control Node in Ansible?

The Control Node is the main computer where Ansible is installed and used to manage the other machines. It sends commands and playbooks to the managed nodes.

2. Why is an inventory file required?

An inventory file tells Ansible which machines are available to manage and provides their network information, such as their IP addresses. It also allows hosts to be organized into groups.

3. What is the purpose of a metagroup?

A metagroup combines multiple groups into one logical group. This allows Ansible commands or playbooks to be applied to all hosts belonging to those groups at once.

4. How do variables simplify Ansible administration?

Variables allow information such as the operating system and username to be assigned to hosts or groups. This makes the inventory easier to manage and allows playbooks to use the assigned information automatically.

5. What is the difference between an ad hoc command and a playbook?

An ad hoc command is used to perform a quick, single task directly on managed nodes. A playbook is a YAML file containing organized tasks that can be executed together to automate multiple operations.

Conclusion/Learnings

Through this laboratory, I learned how to install and verify Ansible, create an inventory, organize managed nodes using groups, and use variables. I also learned how to use ad hoc commands and create playbooks to automate tasks on managed nodes. Overall, the activity helped me understand how Ansible can simplify server management by allowing multiple machines to be managed from a single Control Node.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: N/A

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: N/A

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: N/A

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